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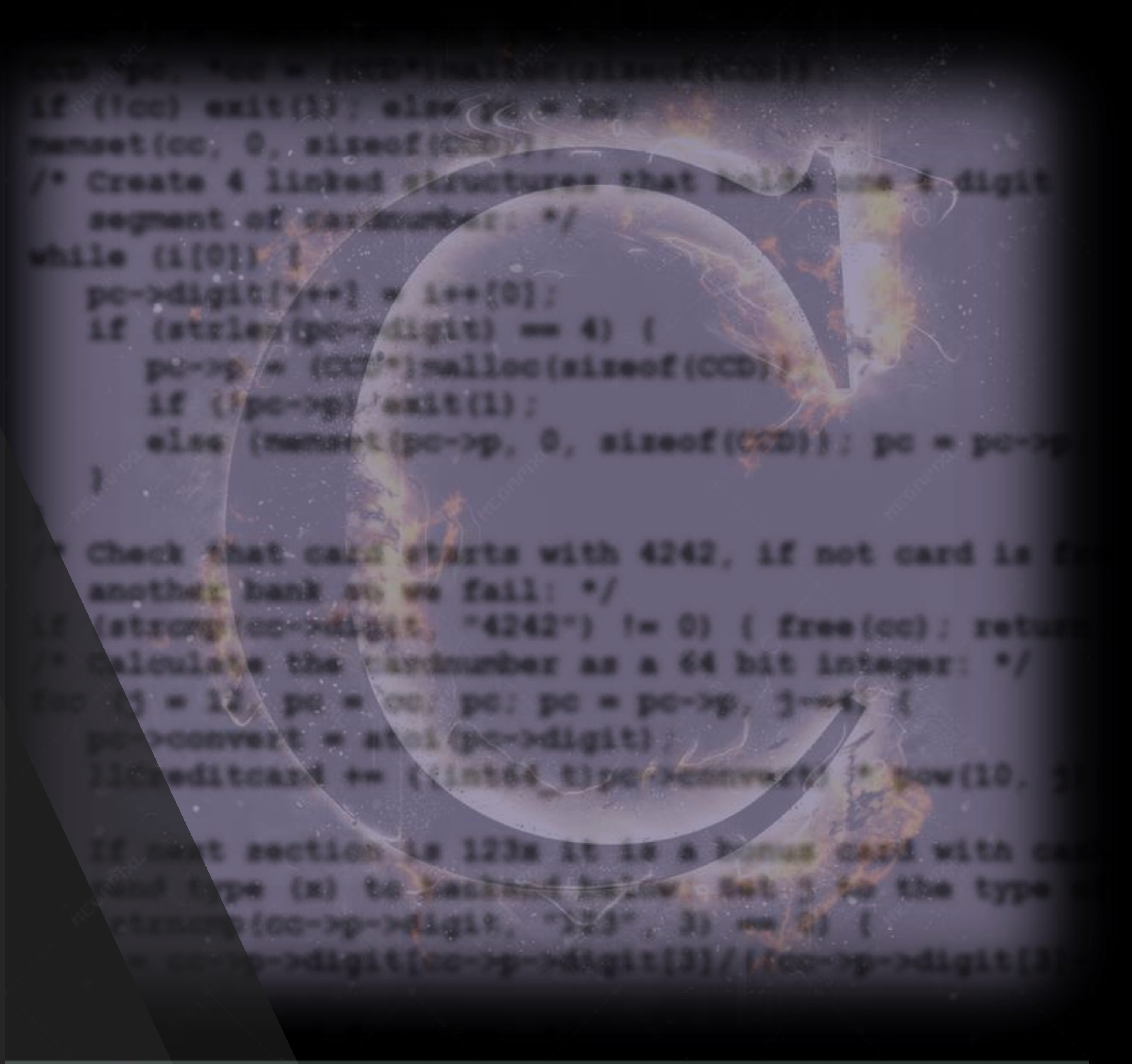
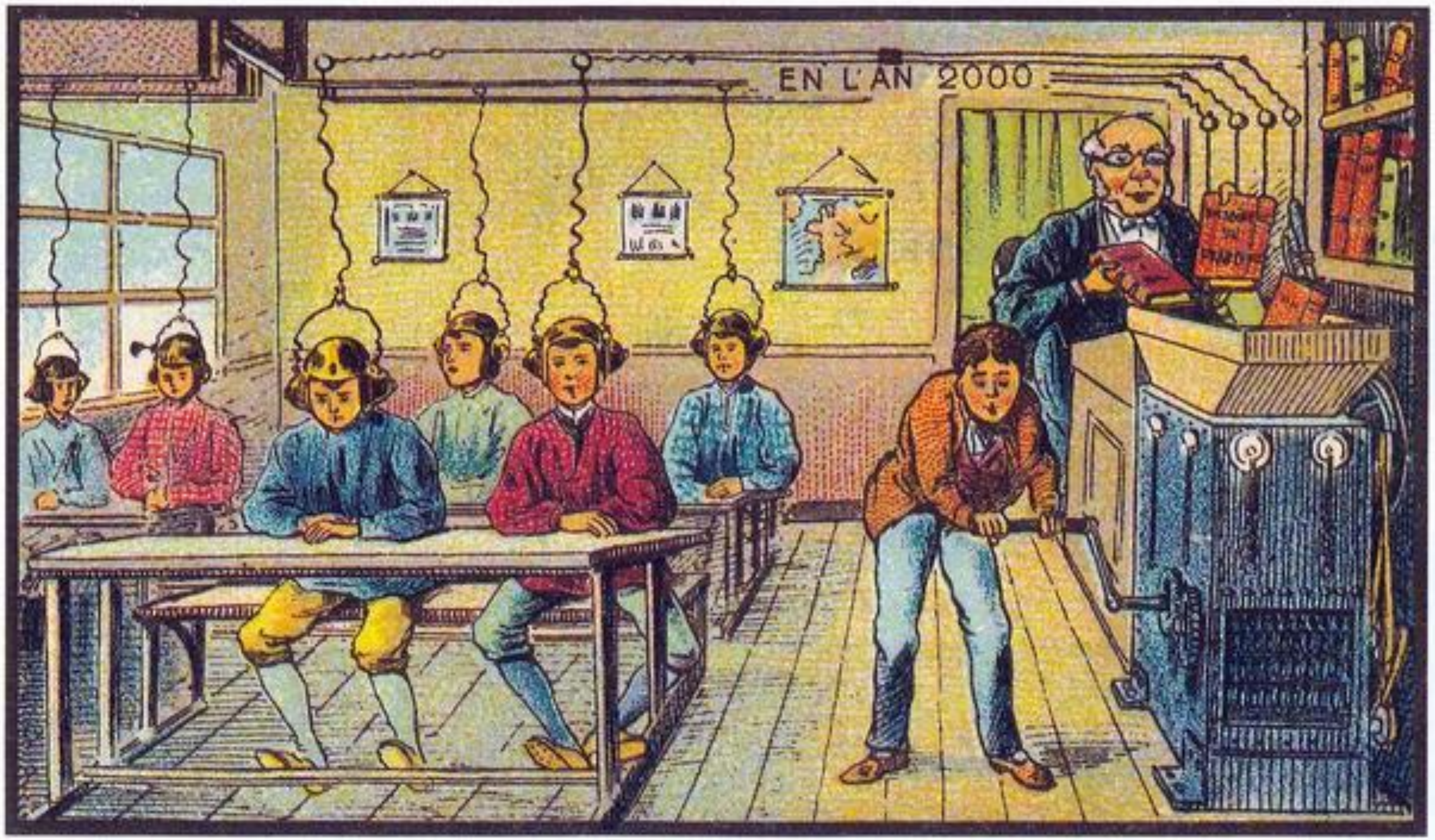


PG3401 C Programmering for Linux

Forelesning 2 (week 03)

Linux and Tools

I apologize for the repetition and info-dump

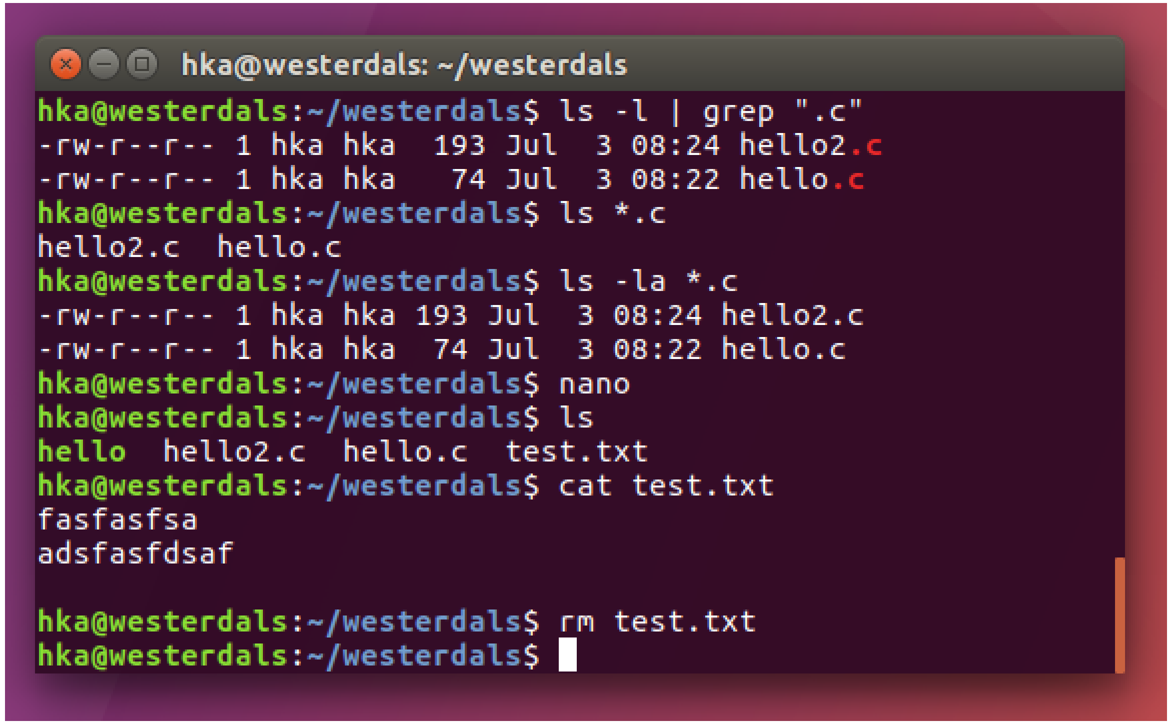


At School

Operating System

- Linux, Windows, OSx, iOs, SintranIII, MS-DOS…
    - File Management
    - Process Management
    - Processing
    - Self organization
    - Handling hardware

To be good at Linux you need to be good with terminal emulators



“Shell” is way to interact with OS

Most common:

- BASH [Bourne Again SHell]
- CSH [C SHell]
- GUI available but we live in denial!

«Commands» in terminal

Terminal navigation

- “TAB” – smart completion
- “UP Arrow/Down Arrow” – previous commands
- “Enter/Return” – Submit the text [commands] into the shell
- “Ctrl – R” – search history of commands
- “Ctrl – R” – Keep searching!

Terminal Interaction

- End command with & to start as separate process.
- **gedit griseenkelt.c &**

- Ctrl-C - to kill a program
- Ctrl-Z – suspend a program
  - Can be restarted with “fg” or “bg”
- Highlight text with mouse
- Auto clipboard
- Middle click to paste

Everything is a file!

- All resources are files
  - I/O devices
  - disks
- Even directories

Unix File Structure

- Everything starts with ‘/’ – root folder
- Typically a user apart from root cannot do much in that folder
- The entire system is well-organized in folders in ‘/’
- (Virtual links)

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Folders

- /bin – Command line binaries
- /boot – files needed for boot
- /dev - devices, hardware
- /etc – configuration files
- **/home** – home directories for users (~/ == /home/hka/)
- /lib – shared libraries for the entire system
- /media – removable disks – could be in mnt
- /mnt – mounted devices
- /opt – optional stuff, used rarely
- /proc – virtual files for processes and kernel info
- /root – root user’s home
- /sbin – system binaries
- /tmp – cleared on every shutdown
- /usr – has pretty much everything
- /var – variable files printed and changed frequently

- (links…)

man

- Manual program
- Provides documentation
- “man PROGRAM” provides documentation about PROGRAM
    - man cd
    - man ls
    - man gcc
- Read man man ☺

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cd

NOTE: On Linux you need a space between cd and ..

- Change directory – “cd”
- With full path
  - cd /home/hka/westerdals
- With user home
  - cd ~/westerdals
- “..” is parent directory
- “.” is current directory
- “cd –” switch to previous directory (like “back” in browser)

ls

- List
  - list all files in the directory
- **ls /home/** -> list files in /home
- **ls –l** list all files in long format[a lot of info](ll in some systems)
- **ls –a** list all files[including the hidden ones]
- Hide a file with '.' as prefix
- (**dir** synonym for **ls**)

cat

- concatenate files and print to standard output
- Usage :
  - “cat file1 file2” will print file1 and file2 next to each other
  - “cat file1” will print the contents of file1
- Usually good to join files by simply placing them together
- Good to use with “split” for network

less/more

- Both are based on vi
- It just displays text
- Navigation
  - space – next page
  - enter – next line
  - page up – prev page
  - Arrow keys up/down == one line up/down
  - q - quit

File commands

- mkdir – create directory
- cp - copy
- mv - move
- rm - remove

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rm

- **rm** – remove
- Usage :
  - **rm file** – deletes file
  - **rm –r dir** – deletes directory with a confirmation prompt
  - **rm –f file** – skips the confirmation prompt
  - **rm –rf** – Oops! Is not the correct reaction after asking for it

You can go “sudo rm –rf /” - but you shouldn’t unless you want to install your system again…

tar

- An archiver/extractor tool
- Usage :
  - **tar cf archive.tar files/directory**
  - **tar xf archive.tar** untars
  - **tar tvf archive.tar** lists files to terminal
- c – create
- v – verbose
- x – extract
- t – list files
- f – indicates the next argument is a compressed file [usually the case]
- z – compression and extraction using gzip

ps

- ps – processes
  - Prints a list of all processes
  - Without any arguments prints only the ones in the current terminal
- ps ax – prints other processes with their paths
- ps ex – prints all parameters

- pid is the process id – displayed in the first column

kill/ killall

- kill pid – exit process with id pid
- kill -9 pid – kill a process with id pid
- killall name – kill all with matching name

- (Make sure you kill the right processes, else you might disturb functionality of the system)

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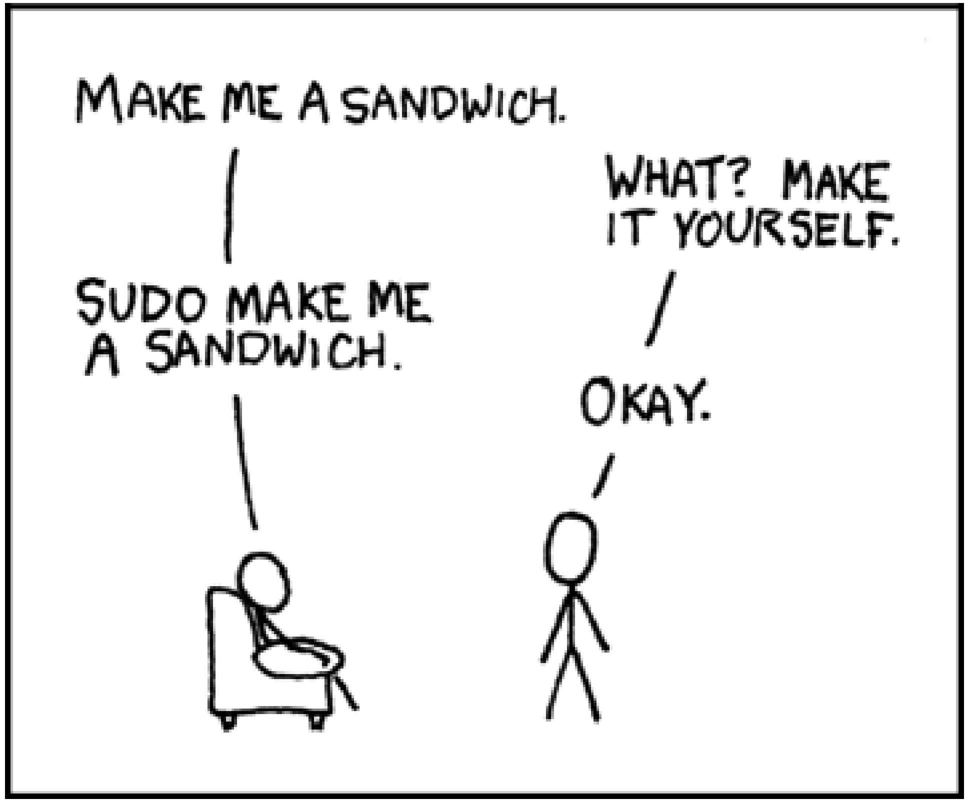
su/sudo

- Super user – root or admin
- Type "su –" makes you login as root
- sudo – "super user" + do
- sudo – for just one command
- "su" sometimes needs "sudo".

- If su doesn't work: Use **sudo –i**)



If something fails, you can just write sudo in-front of it. Remember that when we start with network programming (you need sudo to listen on ports below 1024...)

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grep

- grep – globally search a regular expression and print
- Prints all matching lines
- Regular expression – [aA]nt – searches for ant, Ant
- Usage :
  - grep regex file/files
  - grep “hello” *.txt
  - grep “[hH]ello” *.txt
  - grep –C 10 “hello” *.txt
  - grep –i “hello” *.txt

Permissions

- Permissions are important in Unix based systems
- Three sets:
  - User
  - Group
  - Other
- Three permissions per set
  - Read- r
  - Write - w
  - Execute – x
- Directory flag

chmod

- Change the permissions
- User: **u**(ser) **g**(roup) **o**(others) **a**(all)
- Permissions:
- '+' '-' '='
- r (ead) w (rite) (e) x (e_ecute)

- **chmod uo +r *file***
- **chmod +x *nonexecutable***

- Can also set permissions in octal.

chown

- Change owner
- Usage :

chown *owner file*

wget

- Wget – web Get
- It is useful to grab files across HTTP, FTP
- Usage :

wget **http://www.google.com**
wget **https://www.eastwill.no/pg3401/linux.txt**
wget **https://www.eastwill.no/pg3401/tools.txt**

pipe

- User symbol '|'
- Sending output from one program as input to another:

- **ls - lR | less**
  - Sending file list for less, to see a page at a time
- **ls –l | grep “.txt”**
  - Prints lines containing the text “.txt“ (**ls –l *.txt**)
- Can be combined in series:
- **ls - lR | grep “.txt” | less**

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nano / gedit

Nano:

- The most available editor on Unix platforms
- It is simple, intuitive and not-so-powerful
- Usage :
  - **nano** *filename*
- Ctrl o to save the file
- Ctrl x to exit

We use gedit (when we can), nano is only for emergencies...

Gedit:

- Gedit is more usable, has “codified” text colors
- sudo apt-get install gedit

mount

- "Mount" - sets up a directory to represent the contents of a device. Requires root, unless you want to mount is already in the "/etc/fstab"
- **mount <options> *device folder***
- Mounts "device" in the "folder"
- Common option is "-t" for filesystem type
- Most mount is automatic in modern Linux

Filesystems

- "Ext3" or "ext4": default linux
- "Ntfs" newer Windows
- "VFAT" older windows / dos. Often used on USB-sticks
- "Smbfs" windows shares
- "Iso9660" cd-roms
- "Vboxsf" virtual box

Redirect < >

- Uses symbols '<' and '>'
- Sending data to or from file
- **ls –lR > file.txt**

    output of ls to file.txt

- **rm - rf / 2>&1>/dev/null**

    Sending both standard and error messages to the black hole (useful in scripts).

- **patch -p0 <diff.txt**

    Patch files with diff from diff.txt

Symbolic link

- Windows has shortcuts
- Links are much more *)
- Link to a real file from a "virtual"
- Most programs will read the file it points to
- etc., cp, rm will only operate on the link
- Usage :

  ln –s *target link-name*

*) NTFS har akkurat samme muligheter for symbolske (og harde) linker som Linux Ext3/Ext4 men OS'et bruker det ikke slik som det brukes i systemkatalogene på Linux slik at «ls –lR /» nesten aldri tar slutt.

More commands

- **du** – Disk usage
- **who** – who is logged in
- **free** – displays memory usage
- **top** – displays Linux processes and the resources consumed
- **file** – identify file.

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Summary

- echo “Linux is awesome!”
- General usage of OS for development
- Familiarity with using terminal
- “man” is your friend
- Scripting…

Summary

wget http://www.eastwill.no/pg3401/linux.txt

Work through it?

If you had troubles, this might give you some hints:

wget http://www.eastwill.no/pg3401/linux_with_hints.txt

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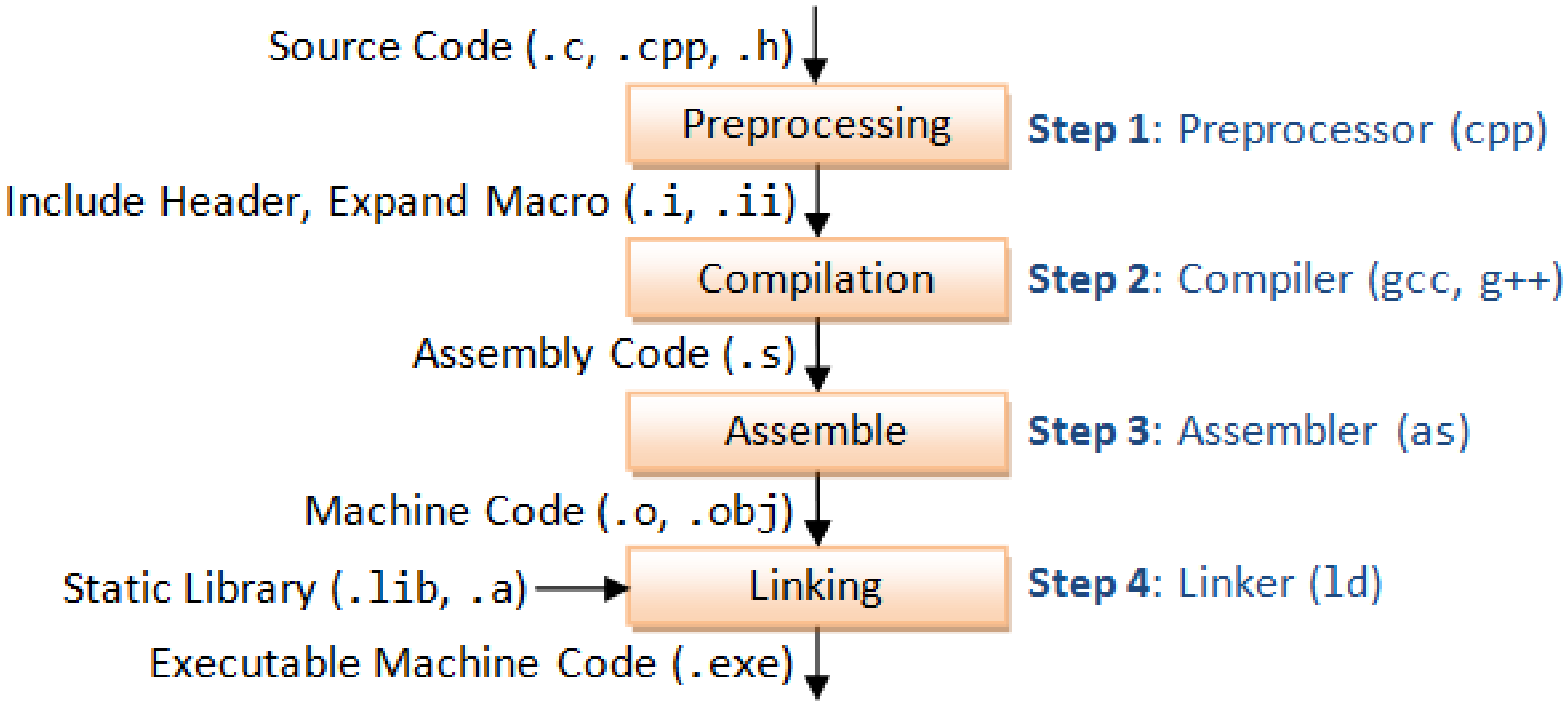
Compiling tools and Debugging tools

Life of C



Source Code (.c, .cpp, .h)
Preprocessing
Step 1: Preprocessor (cpp)
Include Header, Expand Macro (.i, .ii)
Compilation
Step 2: Compiler (gcc, g++)
Assembly Code (.s)
Assemble
Step 3: Assembler (as)
Machine Code (.o, .obj)
Static Library (.lib, .a)
Linking
Step 4: Linker (ld)
Executable Machine Code (.exe)

Editor

- The first human input
- Several editors:
  - Vim
  - Emacs
  - Gedit
  - Nano
  - Notepad
  - …
- Simple text files with extension .c/.h

Recommend everyone use **gedit**, nano is bad, vim replaces sometimes with special UTF8 characters, and emacs is too elite…

Source code

```c
#define MEMSIZE 10
int main(int argc, char *argv[])
{
        //My first buggy program
        printf("Entering main\n");
        int *array;
        int alpha, beta;
        array = (int*)malloc(MEMSIZE*sizeof(int));
        array[5] = array[2]+10;
        alpha = MEMSIZE + 20;
        beta = alpha + MEMSIZE + 30;
}
```

Preprocessor

- Handles all preprocessing
- Preprocessing:
  - All includes – header, sources
  - Macros
- Outputs text file
- Ex : gcc –E in.c –o out.c

After Preprocessing

```
# 1 "test.c"
# 1 "<command-line>"
# 1 "/usr/include/stdc-predef.h" 1 3 4
# 30 "/usr/include/stdc-predef.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/predefs.h" 1 3 4
# 31 "/usr/include/stdc-predef.h" 2 3 4
# 1 "<command-line>" 2
# 1 "test.c"


int main(int argc, char *argv[]){

  printf("Entering main\n");
  int *array;
  int alpha, beta;
  array = (int *)malloc(10*sizeof(int));
  array[5] = array[2]+10;
  alpha = 10 + 20;
  beta = alpha + 10 + 30;
}
```

Compiler

- Conversion from Source to assembly code
- Compiler optimizations
- Ex : **gcc –O0 –S in.c –o out.s**
- With optimization : **gcc –O2 –S in.c –o out.s**

After Compiling- no opt

```
        .file       "pre.c"
        .section    .rodata
.LC0:
        .string     "Entering main"
        .text
        .globl      main
        .type       main, @function
main:
.LFB0:
        .cfi_startproc
        pushq       %rbp
        .cfi_def_cfa_offset 16
        .cfi_offset 6, -16
        movq        %rsp, %rbp
        .cfi_def_cfa_register 6
        subq        $32, %rsp
        movl        %edi, -20(%rbp)
        movq        %rsi, -32(%rbp)
        movl        $.LC0, %edi
        call        puts
        movl        $40, %edi
        call        malloc
        call        malloc
        movq        %rax, -8(%rbp)
        movq        -8(%rbp), %rax
        addq        $20, %rax
        movq        -8(%rbp), %rdx
        addq        $8, %rdx
        movl        (%rdx), %edx
        addl        $10, %edx
        movl        %edx, (%rax)
        movl        $30, -16(%rbp)
        movl        -16(%rbp), %eax
        addl        $40, %eax
        movl        %eax, -12(%rbp)
        leave
        .cfi_def_cfa 7, 8
        ret
        .cfi_endproc
.LFE0:
        .size       main, .-main
        .ident      "GCC: (Ubuntu/Linaro 4.8.1-10ubuntu9) 4.8.1"
        .section    .note.GNU-stack,"",@progbits
```

After Compiling - Opt

```
        .file   "pre.c"
        .section        .rodata.str1.1,"aMS",@progbits,1
.LC0:
        .string "Entering main"
        .section        .text.startup,"ax",@progbits
        .p2align 4,,15
        .globl  main
        .type   main, @function
main:
.LFB0:
        .cfi_startproc
        movl    $.LC0, %edi
        jmp     puts
        .cfi_endproc
.LFE0:
        .size   main, .-main
        .ident  "GCC: (Ubuntu/Linaro 4.8.1-10ubuntu9) 4.8.1"
        .section        .note.GNU-stack,"",@progbits
```

Assembler

- Convert Assembly code to binary code
- Usually called object file
- Non-executable
- Contains references to external libraries
- Output not quite human friendly
- Ex: as in.s –o out.o

- Needing to code in assembler is very rare today; x86 assembler can be used for some special protected calls in kernel drivers, and proprietary assembler might be needed for some (rare) embedded programming.
- But this is 2026, so we can safely learn a programming language from 1989.

Linker

- Links the references to external libraries
- External functions: printf, malloc
- Ex: ld, collect2
- Ex: ld –o executable objectfile –all_libraries

In summary

```
COLLECT_GCC=gcc
COLLECT_LTO_WRAPPER=/usr/lib/gcc/x86_64-linux-gnu/4.8/lto-wrapper
Target: x86_64-linux-gnu
Configured with: ../src/configure -v --with-pkgversion='Ubuntu/Linaro 4.8.1-10ubuntu9' --with-bugurl=file:///usr/share/doc/gcc-4.8/README.Bugs --enable-languages=c,c++,java,go,d,fortran,objc,
obj-c++ --prefix=/usr --program-suffix=-4.8 --enable-shared --enable-linker-build-id --libexecdir=/usr/lib --without-included-gettext --enable-threads=posix --with-gxx-include-dir=/usr/includ
e/c++/4.8 --libdir=/usr/lib --enable-nls --with-sysroot=/ --enable-clocale=gnu --enable-libstdcxx-debug --enable-libstdcxx-time=yes --enable-gnu-unique-object --enable-plugin --with-system-zl
ib --disable-browser-plugin --enable-java-awt=gtk --enable-gtk-cairo --with-java-home=/usr/lib/jvm/java-1.5.0-gcj-4.8-amd64/jre --enable-java-home --with-jvm-root-dir=/usr/lib/jvm/java-1.5.0-
gcj-4.8-amd64 --with-jvm-jar-dir=/usr/lib/jvm-exports/java-1.5.0-gcj-4.8-amd64 --with-arch-directory=amd64 --with-ecj-jar=/usr/share/java/eclipse-ecj.jar --enable-objc-gc --enable-multiarch -
-disable-werror --with-arch-32=i686 --with-abi=m64 --with-multilib-list=m32,m64,mx32 --with-tune=generic --enable-checking=release --build=x86_64-linux-gnu --host=x86_64-linux-gnu --target=x8
6_64-linux-gnu
Thread model: posix
gcc version 4.8.1 (Ubuntu/Linaro 4.8.1-10ubuntu9)
COLLECT_GCC_OPTIONS='-v' '-O2' '-o' 'test' '-mtune=generic' '-march=x86-64'
 /usr/lib/gcc/x86_64-linux-gnu/4.8/cc1 -quiet -v -imultiarch x86_64-linux-gnu test.c -quiet -dumpbase test.c -mtune=generic -march=x86-64 -auxbase test -O2 -version -fstack-protector -Wformat
 -Wformat-security -o /tmp/ccLJWfWn.s
GNU C (Ubuntu/Linaro 4.8.1-10ubuntu9) version 4.8.1 (x86_64-linux-gnu)
        compiled by GNU C version 4.8.1, GMP version 5.1.2, MPFR version 3.1.1-p2, MPC version 1.0.1
GGC heuristics: --param ggc-min-expand=100 --param ggc-min-heapsize=131072
ignoring nonexistent directory "/usr/local/include/x86_64-linux-gnu"
ignoring nonexistent directory "/usr/lib/gcc/x86_64-linux-gnu/4.8/../../../../x86_64-linux-gnu/include"
#include "..." search starts here:
#include <...> search starts here:
 /usr/lib/gcc/x86_64-linux-gnu/4.8/include
 /usr/local/include
 /usr/lib/gcc/x86_64-linux-gnu/4.8/include-fixed
 /usr/include/x86_64-linux-gnu
 /usr/include
End of search list.
GNU C (Ubuntu/Linaro 4.8.1-10ubuntu9) version 4.8.1 (x86_64-linux-gnu)
        compiled by GNU C version 4.8.1, GMP version 5.1.2, MPFR version 3.1.1-p2, MPC version 1.0.1
GGC heuristics: --param ggc-min-expand=100 --param ggc-min-heapsize=131072
Compiler executable checksum: b946adc090b5605f865521be8b22959c
test.c: In function ‘main’:
test.c:5:3: warning: incompatible implicit declaration of built-in function ‘printf’ [enabled by default]
   printf("Entering main\n");
   ^
test.c:8:18: warning: incompatible implicit declaration of built-in function ‘malloc’ [enabled by default]
   array = (int *)malloc(MEMSIZE*sizeof(int));
                  ^
COLLECT_GCC_OPTIONS='-v' '-O2' '-o' 'test' '-mtune=generic' '-march=x86-64'
 as -v --64 -o /tmp/cc3o1HBq.o /tmp/ccLJWfWn.s
GNU assembler version 2.23.52 (x86_64-linux-gnu) using BFD version (GNU Binutils for Ubuntu) 2.23.52.20130913
COMPILER_PATH=/usr/lib/gcc/x86_64-linux-gnu/4.8/:/usr/lib/gcc/x86_64-linux-gnu/4.8/:/usr/lib/gcc/x86_64-linux-gnu/:/usr/lib/gcc/x86_64-linux-gnu/4.8/:/usr/lib/gcc/x86_64-linux-gnu/
LIBRARY_PATH=/usr/local/cuda-5.5/lib64/../lib/:/usr/lib/gcc/x86_64-linux-gnu/4.8/:/usr/lib/gcc/x86_64-linux-gnu/4.8/../../../x86_64-linux-gnu/:/usr/lib/gcc/x86_64-linux-gnu/4.8/../../../../li
b/:/lib/x86_64-linux-gnu/:/lib/../lib/:/usr/lib/x86_64-linux-gnu/:/usr/lib/../lib/:/usr/local/cuda-5.5/lib64/:./:/usr/lib/gcc/x86_64-linux-gnu/4.8/../../../:/lib/:/usr/lib/
COLLECT_GCC_OPTIONS='-v' '-O2' '-o' 'test' '-mtune=generic' '-march=x86-64'
 /usr/lib/gcc/x86_64-linux-gnu/4.8/collect2 --sysroot=/ --build-id --eh-frame-hdr -m elf_x86_64 --hash-style=gnu --as-needed -dynamic-linker /lib64/ld-linux-x86-64.so.2 -z relro -o test /usr/
lib/gcc/x86_64-linux-gnu/4.8/../../../x86_64-linux-gnu/crt1.o /usr/lib/gcc/x86_64-linux-gnu/4.8/../../../x86_64-linux-gnu/crti.o /usr/lib/gcc/x86_64-linux-gnu/4.8/crtbegin.o -L/usr/local/cuda
-5.5/lib64/../lib -L/usr/lib/gcc/x86_64-linux-gnu/4.8 -L/usr/lib/gcc/x86_64-linux-gnu/4.8/../../../x86_64-linux-gnu -L/usr/lib/gcc/x86_64-linux-gnu/4.8/../../../../lib -L/lib/x86_64-linux-gnu
 -L/lib/../lib -L/usr/lib/x86_64-linux-gnu -L/usr/lib/../lib -L/usr/local/cuda-5.5/lib64 -L. -L/usr/lib/gcc/x86_64-linux-gnu/4.8/../../.. /tmp/cc3o1HBq.o -lgcc --as-needed -lgcc_s --no-as-nee
ded -lc -lgcc --as-needed -lgcc_s --no-as-needed /usr/lib/gcc/x86_64-linux-gnu/4.8/crtend.o /usr/lib/gcc/x86_64-linux-gnu/4.8/../../../x86_64-linux-gnu/crtn.o
```

Or

gcc –O2 griseenkelt.c –o griseenkelt

“make” your life easier

- Building tools…

- **make**
- cmake – Not strictly building tool - Powerful

Make

- Classic and still widely used for Unix based platforms (and Windows!)

- **gcc –O2 griseenkelt.c –o griseenkelt**

```
#
# Simple makefile for compiling and linking one single source file
#

griseenkelt : griseenkelt.c
	gcc -O2 griseenkelt.c -o griseenkelt
```

Make

```makefile
#
# Simple makefile for compiling and linking one single source file
#

TARGET = griseenkelt

$(TARGET) : $(TARGET).c
	gcc -O2 $^ -o $@
```

Make

- A bit more complex …

```
#
# Generic makefile for compiling and linking more than one source file
#

OBJS = hello.o number.o # List object files here
# DEPS = number.h
CFLAGS = -O2

%.o: %.c $(DEPS)
	gcc -c -o $@ $< $(CFLAGS)

hello: $(OBJS)
	gcc -o $@ $^ $(CFLAGS)

.phony: clean
clean:
	rm -f *.o
```

Make

```
INCLDIR = ./include
CC = gcc
CFLAGS = -O2
CFLAGS += -I$(INCLDIR)

OBJDIR = obj

_DEPS = number.h
DEPS = $(patsubst %,$(INCLDIR)/%,$(_DEPS))

_OBJS = hello.o number.o
OBJS = $(patsubst %,$(OBJDIR)/%,$(_OBJS))


$(OBJDIR)/%.o: %.c $(DEPS)
	$(CC) -c -o $@ $< $(CFLAGS)

hello: $(OBJS)
	gcc -o $@ $^ $(CFLAGS)

.PHONY: clean
clean:
	rm -f $(OBJDIR)/*.o *~ core $(INCLDIR)/*~
```

cmake

- Cross-platform make
- Generates build scripts that you can use to build
- Really handy when you want to compile across different platforms

```
CMAKE_MINIMUM_REQUIRED( VERSION 2.6 )
PROJECT( Hello )
INCLUDE_DIRECTORIES( include )
ADD_EXECUTABLE( hello hello.c )
TARGET_LINK_LIBRARIES( somelibrary m )
```

Other useful tools

- Gdb – A powerful debugging tool
- Valgrind – Powerful memory tool
- Make sure you compile with –O0[No optimization] and –g option for better results from these tools

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GDB

- GNU Debugger
- Useful to track the errors
- Controlled run
- Display memories

Running code in a debugger might help you, I don't use that as much as I probably should do :-)

GDB

- To run the debugger
  - $ gdb *filename* (or simply) gdb
- Inside gdb environment
  - run (or) r to run the program
  - start / step / stepi
  - bt – back trace the stack
  - print/info/display – to see useful information

GDB output

Starting program: /home/vamsidhar/pg3400/./a.out

```
Program received signal SIGSEGV, Segmentation fault.
0x00000000004005d6 in main (argc=1, argv=0x7fffffffde08) at testdb.c:11
11        a[10] = 'a';
```

Valgrind

- Memory debugging
- Memory leaks
- Uninitialized memory
- Reading/writing to invalid memory
  - Freed memory
  - Out of allocated blocks

Before you finalize your code (commit to company's repo – or submit your exam), always run valgrind!

The best exams usually include valgrind reports for all the tasks.

Valgrind

- To use valgrind:

  valgrind --leak-check=yes --track-origins=yes *filename*

- Overwhelming information!!

Valgrind output

```
==6461== Memcheck, a memory error detector
==6461== Copyright (C) 2002-2012, and GNU GPL'd, by Julian Seward et al.
==6461== Using Valgrind-3.8.1 and LibVEX; rerun with -h for copyright info
==6461== Command: ./a.out
==6461==
==6461== Invalid write of size 4
==6461==    at 0x4005B7: main (testdb.c:8)
==6461==  Address 0x51fc1d0 is not stack'd, malloc'd or (recently) free'd
==6461==
==6461== Invalid read of size 4
==6461==    at 0x4005BD: main (testdb.c:9)
==6461==  Address 0x51fc1d0 is not stack'd, malloc'd or (recently) free'd
==6461==
alpha is 0
==6461==
==6461== HEAP SUMMARY:
==6461==     in use at exit: 40 bytes in 1 blocks
==6461==   total heap usage: 1 allocs, 0 frees, 40 bytes allocated
==6461==
==6461== LEAK SUMMARY:
==6461==    definitely lost: 40 bytes in 1 blocks
==6461==    indirectly lost: 0 bytes in 0 blocks
==6461==      possibly lost: 0 bytes in 0 blocks
==6461==    still reachable: 0 bytes in 0 blocks
==6461==         suppressed: 0 bytes in 0 blocks
==6461== Rerun with --leak-check=full to see details of leaked memory
==6461==
==6461== For counts of detected and suppressed errors, rerun with: -v
==6461== ERROR SUMMARY: 2 errors from 2 contexts (suppressed: 2 from 2)
```

Start using valgrind now when you compile very small programs, helps you learn how output will look :-)

Summary

- Compilation tools : gcc
  - Makefile, Cmake
- Debugging tools
  - gdb – execution bugs
  - valgrind – memory bugs

Lab exercises

- **Help each other becoming used to Linux. Some of you might have used it before, some are new to this OS. Make sure you all help each other – this is one of the most difficult courses you will take at this school, everyone of you will need help from the class! :-) You must attend Campus trainings, trying to learn it all your self will most often fail, hard…**

- **After THIS week everyone in the class MUST have a running Linux environment and be able to edit and compile code**

- **Remember that you need to attend lab exercises physically on campus, writing code WITH your course instructor is the only way to learn :-)**

Høyskolen
Kristiania